

Representation of the data

Given a raw data set, we can rearrange it in two different ways.

❖ **Frequency distribution or Discrete frequency distribution:**

Using the frequency of the variable we can arrange it. This representation of the data is known as frequency distribution.

❖ **Grouped frequency distribution or Continuous frequency distribution:**

Again we can arrange it for the class intervals. For this situation, it is called as Grouped frequency distribution of the variable.

Frequency distribution of ungrouped data:

Given below are marks obtained by 20 students in Math out of 25.

21, 23, 19, 17, 12, 15, 15, 17, 17, 19, 23, 23, 21, 23, 25, 25, 21, 19, 19, 19

Marks Obtained	Tally Marks	Frequency
12	I	1
15	II	2
17	III	3
19	IIII	5
21	III	3
23	IIII	4
25	II	2

Frequency distribution of grouped data:

The presentation of the above data can be expressed into groups. These groups are called classes or the **class interval**.

Each class interval is bounded by two figures called the **class limits**.

Marks	Number of Students (Frequency)
0 - 10	0
10 - 20	11
20 - 30	9

Note: The lower value of a class interval is called lower limit and upper value of that class interval is called the upper limit. Thus, each class interval has lower and upper limits.

For Example:

In the class interval 10 - 20, 10 is the lower limit and 20 is the upper limit.

Exclusive form of data:

The previous table is expressed in the exclusive form.
The class intervals are 0 - 10, 10 - 20, 20 - 30.
In this, we include lower limit but exclude upper limit.
So, 10 - 20 means values from 10 and more but less than 20.
20 - 30 would mean values from 20 and more but less than 30.

Data in the inclusive form:

Marks obtained by 20 students in Math test are given below.
23, 0, 14, 10, 15, 3, 8, 16, 18, 20, 1, 3, 20, 23, 24, 15, 24, 22, 14, 13
Let us represent this data in the inclusive form.

Marks	Number of Students (Frequency)
0 - 10	6
11 - 20	9
21 - 30	5

How to make it continuous?

If “ d ” is the gap between the upper limit of any class and the lower limit of the succeeding class, the class boundaries for any class are then given by:

$$\begin{aligned}\text{Upper class boundary} &= \text{Upper class limit} + \frac{d}{2} \\ \text{Lower class boundary} &= \text{Lower class limit} - \frac{d}{2}\end{aligned}$$

Marks	Number of Students (Frequency)
-0.5 – 10.5	6
10.5 – 20.5	9
20.5 – 30.5	5

Summarizing a raw data set or an organized data set

There are two basic properties of a quantitative data set that are commonly studied. These are central tendency and variability (or dispersion).

Central Tendency:

Quite often it is found that the entries in data set cluster around a central (or middle) value. This behavior of the data set is called the central tendency. The main Challenge is to locate a central value around which the clustering takes place. Three standard methods to measure the location of central tendency are:

- Mean
- Median
- Mode

Variability or Dispersion:

Variability or dispersion of data set means the amount of discrepancies among the data entries. There are several ways to measure dispersion or variability in a data set and these are:

- Range
- Quartile deviation
- Variance
- Standard deviation

Arithmetic Mean or Average

❖ Mean of “ n ” observations $(x_1, x_2, \dots, x_n)$ is given by

$$\bar{x} = \frac{1}{n} \sum x_i .$$

Example: A Monthly income of 10 families in a city is given by:
1600, 1560, 1440, 1530, 1670, 1860, 1750, 1910, 1490, 1800.

Ans:

$$\bar{x} = \frac{1}{10} \sum_{i=1}^{10} x_i = \frac{16610}{10} = 1661.$$

❖ In case of the discrete frequency distribution:

If f_i 's are the frequencies of the variable x_i 's then mean

$$\bar{x} = \frac{1}{N} \sum_{i=1}^n f_i x_i , \quad \text{where } N = \sum_{i=1}^n f_i .$$

Example: The following data represents the marks obtained by 60 students of a class. Obtain the average marks.

Marks	20	30	40	50	60	70
No. of Students	8	12	20	10	6	4

Ans:

Marks (x)	No. of Students (f)	fx
20	8	160
30	12	360
40	20	800
50	10	500
60	6	360
70	4	280
	$N = 60$	$\Sigma fx = 2460$

$$\begin{aligned}
 \bar{x} &= \frac{1}{N} \sum_{i=1}^n f_i x_i \\
 &= \frac{2460}{60} \\
 &= 41
 \end{aligned}$$

❖ In case of the continuous frequency distribution:

If f_i 's are the frequencies of the variable x_i 's then mean

$$\bar{x} = \frac{\sum(\text{midvalue of each class}) \times f_i}{\sum f_i}.$$

Example: Obtain the Arithmetic mean for the following data.

Marks	0-10	10-20	20-30	30-40	40-50	50-60
No. of Students	12	18	27	20	17	6

Ans:

Marks	No. of Students (f)	Mid-point (x)	fx
0-10	12	5	60
10-20	18	15	270
20-30	27	25	675
30-40	20	35	700
40-50	17	45	765
50-60	6	55	330

$$\bar{x} = \frac{2800}{100} = 28$$

Calculation of mean by using deviation concept:

Sometime the values of the variable (x) or frequency (f) or both are large. Then, the calculation of mean by previous formulas is quite time-consuming. Hence, to avoid such situations we are calculating mean by taking the deviations of the given values from any arbitrary point “ A ” as explained below:

Discrete frequency distribution: $\bar{x} = A + \frac{1}{N} \sum_{i=1}^n f_i d_i$,

where “ A ” is an arbitrary point, $d_i = x_i - A$ and $N = \sum_{i=1}^n f_i$.

Continuous frequency distribution: $\bar{x} = A + \frac{h}{N} \sum_{i=1}^n f_i d_i$,

where “ A ” is an arbitrary point, “ h ” is the magnitude of class interval and $N = \sum_{i=1}^n f_i$. Here, $d_i = \frac{x_i - A}{h}$, where x_i ’s are the mid value of each class.

Example:

C.I.	0-8	8-16	16-24	24-32	32-40	40-48
Frequency	8	7	16	24	15	7

range of values: 0 to 48
Though we can choose any A but a reasonable choice would be to take the middle value in highest frequency.

Choose $A = \frac{24+32}{2} = 28$. $h = 8$

Ans:

C.I.	Mid-Value (x)	Frequency (f)	$d = \frac{x - A}{h}$	fd
0-8	4	8	-3	-24
8-16	12	7	-2	-14
16-24	20	16	-1	-16
24-32	28	24	0	0
32-40	36	15	1	15
40-48	44	7	2	14
Total		77		-25

$$\bar{X} = A + \frac{h}{N} \sum_{i=1}^n f_i d_i = 28 + \frac{8 \times (-25)}{77} = 25.404$$

Measure of Central Tendency	Merits	Demerits :
a) Arithmetic Mean	1) It is rigidly defined. 2) It is easy to understand and easy to calculate. 3) It is based on all the observations. 4) It is suitable for further mathematical manipulations. 5) Of all the averages, it is affected least by sampling fluctuations.	1) It is affected very much by extreme values 2) It cannot be found for distributions with open end classes. 3) It cannot be used if measurements are qualitative.

Median

- Median of a distribution is the value of the variable which divides it into two equal parts.
- It is the value which exceeds and is exceeded by the same number of observations. Thus the median is called as a “positional average”.

Evaluation of Median: For ungrouped data,

- (i) Odd number of observations. (middle value after the values has been arranged in ascending or descending order of magnitude)
- (ii) Even number of observations. (arithmetic mean of two middle terms after the values has been arranged in ascending or descending order of magnitude)

Ex: Find the median of the values 25, 20, 15, 35, 18.

Ex: Find the median of the values 8, 20, 50, 25, 15, 30.

Cumulative Frequency

The number of goals scored per match by Kriti during a hockey season was recorded.

Goals	Frequency	Cumulative Frequency
0	1	1 ← scores of 0
1	6	7 ← scores of 0 or 1
2	7	14 ← scores of 0, 1 or 2
3	2	16 ← scores of 0, 1, 2 or 3
4	3	19 ← scores of 0, 1, 2, 3 or 4
5	1	20 ← scores of 0, 1, 2, 3, 4 or 5

(i) Discrete Frequency distribution:

Even: median is the arithmetic mean of $\left(\frac{N}{2}\right)$ th and $\left(\frac{N}{2} + 1\right)$ th observation

Odd: median is the $\frac{N+1}{2}$ th observation

What is the median number of goals scored by Kriti during a game?

Goals	Frequency	Cumulative Frequency
0	1	1 ← scores of 0
1	6	7 ← scores of 0 or 1
2	7	14 ← scores of 0, 1 or 2
3	2	16 ← scores of 0, 1, 2 or 3
4	3	19 ← scores of 0, 1, 2, 3 or 4
5	1	20 ← scores of 0, 1, 2, 3, 4 or 5

There are 20 observations, which is an even number. Therefore, the median is the arithmetic mean of $\left(\frac{20}{2}\right)$ th and $\left(\frac{20}{2} + 1\right)$ th observation = 10th and 11th observation. According to the table, the numbers in the 10th and 11th position are 2's. Therefore, the median is $\frac{2+2}{2} = 2$.

(ii) Continuous Frequency distribution:

In case of continuous frequency distribution, the class corresponding to ^{cumulative frequency} the c.f. just greater than $N/2$ is called the median class and the value of median is obtained by the following formula:

$$\text{Median} = l + \frac{h}{f} \left(\frac{N}{2} - c \right)$$

Where,

l -is the lower limit of the median class,

f -is the frequency of the median class,

h -is the magnitude of the median class,

c -is the c.f. of the class preceding the median class

Note: The median formula can only be used only for continuous classes without any gaps, i.e., for exclusive type classifications.

Example: Find the median of the following data:

Wages (in Rs.)	2000-3000	3000-4000	4000-5000	5000-6000	6000-7000
No. of workers	3	5	20	10	5

Ans:

<i>Wages (in Rs.)</i>	<i>No. of Employees</i>	<i>c.f.</i>
2000-3000	3	3
3000-4000	5	8
4000-5000	20	28
5000-6000	10	38
6000-7000	5	43
	N = 43	

$$\frac{N}{2} = \frac{43}{2} = 21.5$$

Cumulative frequency just greater than 21.5 is 28 and the corresponding class is 4000-5000. Thus the median class is 4000-5000.

$$l = 4000; h = 1000; f = 20; c = 8$$

$$\text{Median} = 4000 + \frac{1000}{20} (21.5 - 8)$$

$$\text{Median} = 4675.$$

Example: Find the value of the median from the following data:

No. of days for which absent (less than)	5	10	15	20	25	30	35	40	45
No. of students	29	224	465	582	634	644	650	653	655

Ans: The given cumulative frequency will be converted into ordinary frequency as follows:

<i>Class-Interval</i>	<i>Cumulative frequency</i>	<i>Ordinary frequency</i>
0-5	29	$29 = 29$
5-10	224	$224 - 29 = 195$
10-15	465	$465 - 224 = 241$
15-20	582	$582 - 465 = 117$
20-25	634	$634 - 582 = 52$
25-30	644	$644 - 634 = 10$
30-35	650	$650 - 644 = 6$
35-40	653	$653 - 650 = 3$
40-45	655	$655 - 653 = 2$

$$N = 655 \Rightarrow \frac{N}{2} = 327.5$$

Median

$$\begin{aligned}
 &= 10 + \frac{\frac{655}{2} - 224}{241} \times 5 \\
 &= 10 + \frac{103.5 \times 5}{241} = 10 + 2.15 \\
 &= 12.15
 \end{aligned}$$

State the merits of Median.

- It is easy to calculate and understand.
- It eliminates the effect of extreme values to variable by which it is not affected.
- It is capable of further algebraic treatment for analysing other measures.

What are the limitations of Median

- For calculating median it is necessary to arrange the data, whereas other averages do not need any arrangement.
- Since median is positional average, it's value is not determined by each and every observation.
- The median in some cases cannot be computed exactly, as in the case of mean.i.e When the number of items included in a series of data is even.
- The value of median is affected more by sampling fluctuations.

Mode

Definition: Mode is the value of the variable which is predominant in the series.

1. For a discrete frequency distribution , mode is the value of x –*corresponding to maximum frequency.*

x	1	2	3	4	5	6	7	8
f	4	9	16	25	22	15	7	3

Mode = 4.

2. For Continuous Frequency distribution: Identify modal class, the class that has the highest frequency.

$$\text{Mode} = l + \frac{h(f_1 - f_0)}{(f_1 - f_0) - (f_2 - f_1)} = l + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2}$$

l —lower limit of the modal class, h —magnitude of the modal class

f_1 -frequency of the modal class

f_0 and f_2 -frequencies of the class preceding and succeeding the modal class

Example: Find mode

<i>C.I.</i>	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
<i>Frequency</i>	5	8	7	12	28	20	10	10

Hence the maximum frequency is 28.

Thus the modal class is 40-50.

$$\text{Mode} = 40 + 10 \frac{(28-12)}{2*28-12-20} = 46.67$$

A distribution is having only one mode is called *Unimodal*.

If it contains more than one mode, it is called *bimodal* or *multimodal*.

Note: In the following three cases, mode can not be obtained by using the above formula:

- (a) When the highest frequency is observed at the beginning of the frequency table.
- (b) When the highest frequency is observed at the ending of the frequency table.
- (c) When two or more class intervals contain the same maximum frequencies.

However, in the above three cases, mode can be obtained by using either a method called ‘Grouping method’ or ‘empirical relationship between arithmetic mean, median and mode.

The empirical relationship between mean, median and mode is

$$\text{Mean} - \text{Mode} = 3 (\text{Mean} - \text{Median})$$

Or,

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean.}$$

Calculate the Mean, Median and Mode for the following data.

<i>Variable</i>	10-13	13-16	16-19	19-22	22-25	25-28	28-31	31-34	34-37	37-40
<i>Frequency</i>	8	15	27	51	75	54	36	18	9	7

Mean=24.19

Median=23.96

Mode=23.6

Uses of Mode :Mode finds an important place in marketing studies, where a manager of a business concern is interested in knowing about the size which has the highest concentration of items. For example, in placing an order for shoes or ready made garments, the model size helps because this size and other sizes around it are in common demand. It is also used in dealing with non-quantitative data

Measure of Dispersion

❖ *Scatteredness (homogeneity or heterogeneity)*

The degree to which numerical data tend to spread about an average value is called variation or dispersion of data.

Measures of Dispersion:

- Range
- Quartile Deviation
- Mean Deviation
- Standard Deviation

Range

Definition: Difference between the value of the smallest item and the value of the largest item in the distribution.

$$\text{Range} = L - S$$

L – Largest Value, S- Smallest Value

The *relative measure* corresponding to range is called the *coefficient of range*,

$$\text{Coefficient of Range} = \frac{L-S}{L+S}$$

Example 1: The following are the prices of shares of a company from Monday to Saturday:

Days	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Price(Rs.)	200	210	208	160	220	250

Calculate the range and its coefficient.

Solution: $\text{Range} = L - S = 250 - 160 = 90$

Range = Rs. 90

$$\text{Coefficient of Range} = \frac{L-S}{L+S} = \frac{250-160}{250+160} = \frac{90}{410} = 0.22$$

In a frequency distribution, range is calculated by taking the difference between the lower limit of the lowest class and the upper limit of the highest class.

Example 2:

<i>Marks</i>	10-20	20-30	30-40	40-50	50-60	60-70
<i>No. of Students</i>	12	18	27	20	17	6

$$\text{Range} = L - S = 70 - 10 = 60$$

$$\text{Coefficient of Range} = \frac{L-S}{L+S} = \frac{70-10}{70+10} = \frac{60}{80} = 0.75$$

Partitions:

These are the values which divide the series into a number of equal parts.

Quartiles: The three points which divide the series into four equal parts are called quartiles. It is denoted by Q_1, Q_2, Q_3 .

Deciles: The nine points which divide the series into ten equal parts are called deciles. It is denoted by $D_1, D_2, \dots, D_9$.

Percentiles: The ninety-nine points which divide the series into hundred equal parts are called percentiles. It is denoted by $P_1, P_2, \dots, P_{99}$.

For discrete frequency distribution:

Quartiles:- Q_k : Find c. f.; Even: Average of $\left(\frac{kN}{4}\right)th$ and $\left(\frac{kN}{4} + 1\right)th$ observation
Odd: $\left(\frac{k(N+1)}{4}\right)th$ observation.

Here, $k = 1, 2, 3$.

Deciles:- D_k : Find c. f.; Even: Average of $\left(\frac{kN}{10}\right)th$ and $\left(\frac{kN}{10} + 1\right)th$ observation
Odd: $\left(\frac{k(N+1)}{10}\right)th$ observation.

Here, $k = 1, 2, 3, 4, 5, 6, 7, 8, 9$.

Percentiles:- P_k : Find c. f.; Even: Average of $\left(\frac{kN}{100}\right)th$ and $\left(\frac{kN}{100} + 1\right)th$ observation
Odd: $\left(\frac{k(N+1)}{100}\right)th$ observation.

Here, $k = 1, 2, \dots, 99$.

For continuous frequency distribution:

Quartiles:- Step-1: Find quartile class by:

Compute $\frac{kN}{4}$; Identify the same value in cf list, otherwise find cf just greater than $\frac{kN}{4}$.

Step-2: Use the formula

$$Q_k = l + \frac{h}{f} \left(\frac{kN}{4} - c \right) ; k = 1, 2, 3.$$

Deciles:- Step-1: Find decile class by:

Compute $\frac{kN}{10}$; Identify the same value in cf list, otherwise find cf just greater than $\frac{kN}{10}$.

Step-2: Use the formula

$$D_k = l + \frac{h}{f} \left(\frac{kN}{10} - c \right) ; k = 1, 2, \dots, 9.$$

Percentiles:- Step-1: Find percentile class by:

Compute $\frac{kN}{100}$; Identify the same value in cf list, otherwise
find cf just greater than $\frac{kN}{100}$.

Step-2: Use the formula

$$P_k = l + \frac{h}{f} \left(\frac{kN}{100} - c \right) ; k = 1, 2, \dots, 99.$$

Quartile Deviation

Definition: Average amount by which the two quartiles differ from the median.

$$\text{Quartile Deviation (Q.D.)} = \frac{Q_3 - Q_1}{2}$$

- The Median $\pm$ Q.D. covers exactly 50 per cent of the observations.
- When Q.D. is very small, it describes high uniformity or small variation of the central 50% items, and a high Q.D. means that the variation among the central items is large.

Relative measure of Q.D.

$$\text{Coefficient of Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

It can be used to compare the degree of variation in different distributions.

Example 3: Calculate the value of Q.D. and its coefficient of Q.D. from the following data.

Roll No.	1	2	3	4	5	6	7
Marks	20	28	40	12	30	15	50

Solution: Marks in ascending order **12 15 20 28 30 40 50**

$Q_1 = \text{Size of } \frac{N+1}{4} \text{th item} = \text{Size of } \frac{7+1}{4} = 2^{\text{nd}} \text{ item.}$
 Size of 2nd item is 15. Hence **$Q_1 = 15$**

$Q_3 = \text{Size of } 3\left(\frac{N+1}{4}\right) \text{th item} = \text{Size of } 3\left(\frac{7+1}{4}\right) = 6^{\text{th}} \text{ item.}$
 Size of 6th item is 40. Hence **$Q_3 = 40$** .

$$\therefore Q.D. = \frac{Q_3 - Q_1}{2} = \frac{40 - 15}{2} = \mathbf{12.5}$$

$$\text{Coefficient of } Q.D. = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{40 - 15}{40 + 15} = \mathbf{0.455}$$

Example 4: Compute the value of Q.D. and its coefficient from the following data.

Marks	10	20	30	40	50	60
No. of Students	4	7	15	8	7	2

Solution:

<i>Marks</i>	<i>Frequency</i>	<i>cumulative frequency</i>
10	4	4
20	7	11
30	15	26
40	8	34
50	7	41
60	2	43

$Q_1 = \text{Size of } \frac{N+1}{4} \text{ th item} = \text{Size of } 11^{\text{th}} \text{ item.}$

Size of 11^{th} item is 20. Hence $Q_1 = 20$

$Q_3 = \text{Size of } 3\left(\frac{N+1}{4}\right) \text{ th item} = \text{Size of } 3\left(\frac{43+1}{4}\right) = 33^{\text{rd}} \text{ item.}$

Size of 33^{rd} item is 40. Hence $Q_3 = 40$.

$$\therefore Q.D. = \frac{Q_3 - Q_1}{2} = \frac{40 - 20}{2} = \mathbf{10}$$

$$\text{Coefficient of } Q.D. = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{40 - 20}{40 + 20} = \mathbf{0.333}$$

Example 5: Compute the value of Q.D. and coefficient of Q.D. from the following data

<i>C.I.</i>	10-20	20-30	30-40	40-50	50-60	60-70	70-80
<i>f</i>	12	19	5	10	9	6	6

Solution:

<i>Marks</i>	<i>Frequency</i>	<i>Cumulative Frequency</i>
10-20	12	12
20-30	19	31
30-40	5	36
40-50	10	46
50-60	9	55
60-70	6	61
70-80	6	67
	<i>N</i> = 67	

$$Q_1 = \text{Size of } \frac{N}{4} \text{th item} = \text{Size of } \frac{67}{4} = 16.75^{\text{th}} \text{ item.}$$

Q_1 lies in the interval **20-30**

$$Q_1 = l + \frac{\frac{N}{4} - c}{f} \times h \quad l = 20, N/4 = 16.75, c = 12, f = 19, h = 10$$

$$Q_1 = 20 + \frac{\frac{67}{4} - 12}{19} \times 10 = 20 + 2.5 = 22.5$$

Hence $Q_1 = 22.5$

$$Q_3 = \text{Size of } \frac{3N}{4} \text{ th item} = \text{Size of } \frac{3 \times 67}{4} = 50.25^{\text{th}} \text{ item.}$$

Q_3 lies in the class **50-60**.

$$Q_3 = l + \frac{\frac{3N}{4} - c}{f} \times h \quad l = 50, 3N/4 = 50.25, c = 46, f = 9, h = 10$$

$$Q_3 = 50 + \frac{50.25 - 46}{9} \times 10 = 50 + 4.72 = 54.72$$

Hence $Q_3 = 54.72$

$$Q.D. = \frac{Q_3 - Q_1}{2} = \frac{54.72 - 22.5}{2} = 16.11$$

$$\text{Coefficient of } Q.D. = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{54.72 - 22.5}{54.72 + 22.5} = 0.4172$$

Merits: When a frequency distribution contains open end classes, Q.D. is the appropriate measure of dispersion.

Demerits: It ignores remaining 50% of the data.

Mean Deviation

Definition: M.D. is the average difference between the observations in a distribution and the *median* or *mean* of that series.

$$\text{Mean Deviation or M.D.} = \frac{\sum |D|}{N}$$

Where $|D|$ is the deviations from *median* ignoring signs.

For individual observations:

- (i) Compute median of the series.
- (ii) Take deviations of items from median ignoring $\pm$ signs and denote these deviations by $|D|$.
- (iii) Obtain the total of these observations, $|D|$.
- (iv) Divide the total obtained in step (iii) by the number of observations to get the value of *mean deviation*.

Relative Measure of M.D. :

$$\text{Co-efficient of M.D.} = \frac{M.D.}{Median}$$

Example 5: Calculate the mean deviation of the two income groups.

I (Rs.)	II (Rs.)
4000	3000
4200	4000
4400	4200
4600	4400
4800	4600
	4800
	5800

Solution:

Mean deviation : **I group** M.D. = $\frac{\sum |D|}{N}$

Median = $\frac{N+1}{2}$ *th* item = $\frac{5+1}{2} = 3^{\text{rd}}$ item. Size of the 3rd item = 4400.

Group I		Group II	
Rs.	D	Rs.	D
4000	400	3000	1400
4200	200	4000	400
4400	0	4200	200
4600	200	4400	0
4800	400	4600	200
		4800	400
		5800	1400
N = 5	$\sum D = 1200$	N = 7	$\sum D = 4000$

$$\text{M.D.} = \frac{1200}{5} = \mathbf{240}$$

i.e., the average deviation of the individual incomes from the median income is **Rs. 240**.

Mean deviation : **II group** $M.D. = \frac{|D|}{N}$

$$\text{Median} = \frac{N+1}{2} \text{th item} = \frac{7+1}{2} = 4^{\text{th}} \text{ item.}$$

Size of the 4th item = 4400.

$$M.D. = \frac{4000}{7} = 571.43$$

i.e., the average deviation of the individual incomes from the median income is **Rs. 571.43**.

$$\text{Co-efficient of M.D. (I - Group)} = \frac{M.D.}{Median} = \frac{240}{4400} = 0.055$$

$$\text{(II- Group)} = \frac{571.43}{4400} = 0.13$$

Mean deviation – Discrete Series

$$\text{M.D.} = \frac{\sum f|D|}{N}$$

- (i) Compute median ^($\tilde{x}$ say) of the series.
- (ii) Take deviations of items from median ignoring $\pm$ signs and denote these deviations by $|D| = |x_i - \tilde{x}|$
- (iii) Multiply these deviations by the respective frequencies and Obtain the total, $\sum f|D|$
- (iv) Divide the total obtained in step (iii) by the number of observations to get the value of *mean deviation*.

Example 6: The number of telephone calls received at an exchange in 245 successive one-minute intervals are shown in the following frequency distribution. Compute the mean deviation about the median. *Also find mean deviation from mean and coefficient of M.D. from mean.*

<i>Number of Calls</i>	0	1	2	3	4	5	6	7
<i>Frequency</i>	14	21	25	43	51	40	39	12

Solution:

<i>No. of Calls</i> (x_i)	f	$c.f.$	$ D = x_i - \tilde{x} $	$f D $
0	14	14	4	56
1	21	35	3	63
2	25	60	2	50
3	43	103	1	43
4	51	154	0	0
5	40	194	1	40
6	39	233	2	78
7	12	245	3	36
	$N = 245$		$\sum f D = 366$	

Median = Size of $\frac{N+1}{2}$ th item = $\frac{245+1}{2} = 123^{\text{rd}}$ item. $\text{Median}(\tilde{x}) = 4.$

Hence the median value is 4.

$$\text{M.D.} = \frac{\sum f|D|}{N} = \frac{366}{245} = 1.49$$

In Continuous Series:

We have to obtain the mid-points of the various classes and take the deviations of these **mid-points** from median.

Calculate the coefficient of mean deviation from the following data:

<i>Marks</i>	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
<i>No. of Students</i>	2	6	12	18	25	20	10	7

Mean deviation = 12.94 (*from median*)

coefficient of mean deviation = 0.236 (*from median*)

Ans:

$$N = 100, \frac{N}{2} = 50,$$

Median class: 50-60

$$\text{Median}(\tilde{x}) = 50 + \frac{10}{25}(50-38) \\ = 54.8$$

$$\text{M.D.} = \frac{\sum f|d|}{N} = \frac{1294.8}{100} \\ = 12.948$$

$$\text{Coef. mean deviation} = \frac{\text{M.D.}}{\text{Median}} \\ = \frac{12.948}{54.8} = 0.236 \\ = 23.6\%$$

Marks C.I.	Frequency f	Mid-value (x)	C. f.	d = x _i - $\tilde{x}$	f d
10-20	2	15	2	39.8	79.6
20-30	6	25	8	29.8	178.8
30-40	12	35	20	19.8	237.6
40-50	18	45	38	9.8	176.4
50-60	25	55	63	0.2	5
60-70	20	65	83	10.2	204
70-80	10	75	93	20.2	202
80-90	7	85	100	30.2	211.4

Exercise. Find the mean deviation from mean and coefficient of M.D. from mean for the above data.

$$\sum f|d| = 1294.8$$

Standard deviation

(i) For the ungrouped data:

$$\text{Variance} = \sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N} = \frac{1}{N} \sum x_i^2 - \left(\frac{1}{N} \sum x_i \right)^2 = \frac{1}{N} \sum x_i^2 - \bar{x}^2$$

(ii) Discrete or Continuous frequency distribution

$$\begin{aligned} \sigma^2 &= \frac{1}{N} \sum_i f_i (x_i - \bar{x})^2 \\ &= \frac{1}{N} \sum f_i x_i^2 - \left(\frac{1}{N} \sum f_i x_i \right)^2 \text{ or } \frac{1}{N} \sum f_i d_i^2 - \left(\frac{1}{N} \sum f_i d_i \right)^2 \end{aligned}$$

(Discrete case with $d_i = x_i - A$)

$$h^2 \left[\frac{1}{N} \sum f_i d_i^2 - \left(\frac{1}{N} \sum f_i d_i \right)^2 \right]$$

(Continuous case with $d_i = \frac{x_i - A}{h}$)

$$\text{Standard deviation} = \sqrt{\text{variance}} \quad \sigma = \sqrt{\frac{1}{N} \sum (x_i - \bar{x})^2} \quad \text{or} \quad \sqrt{\frac{1}{N} \sum f_i (x_i - \bar{x})^2}$$

$\bar{x}$ – Arithmetic mean of the distribution

Recall:

$$\bar{x} = A + \frac{1}{N} \sum_{i=1}^n f_i d_i \text{ (mean for discrete), } \bar{x} = A + \frac{h}{N} \sum_{i=1}^n f_i d_i, \text{ (mean for continuous)}$$

(2) Relations between measures of dispersion

(i) Quartile deviation = $2/3$ (standard deviation)

(ii) Mean deviation = $4/5$ (standard deviation)

Coefficient of Variation

σ - standard deviation

$\bar{x}$ - mean

Coefficient of Variation : C.V. = $\frac{\sigma}{\bar{x}} \times 100$ (Relative Measure)

For comparing the variability of two series, we calculate the co-efficient of variations for each series. The series having greater C.V. is said to be more variable than the other and the series having lesser C.V. is said to be more consistent (or homogenous) than the other.

Qn. Can C.V. = $\frac{\sigma}{\bar{x}}$ be greater than 1?

Check for the data: $\{1, 1, 10\}$. Mean $\bar{x} = \frac{12}{3} = 4$, $\sigma = \sqrt{\frac{(1-4)^2 + (1-4)^2 + (10-4)^2}{3}} = 4.24$

- 3) The score of two players A and B in ten innings during a certain season are:

A	32	28	47	63	71	39	10	60	96	14
B	19	31	48	53	67	90	10	62	40	80

Find which of the two players A, B is more consistent in scoring.

Solution: Calculation of Coefficient of Variation

X	$(X - \bar{X})$	$(X - \bar{X})^2$
32	-14	196
28	-18	324
47	+1	1
63	+17	289
71	+25	625
39	-7	49
10	-36	1296
60	+14	196
96	+50	2500
14	-32	1024
$\sum X = 460$	0	6500

$$\bar{X} = \frac{460}{10} = 46$$
$$\sigma_A^2 = \frac{\sum (x_i - \bar{x})^2}{N} = \frac{6500}{10} = 650$$

Y	$(Y - \bar{Y})$	$(Y - \bar{Y})^2$
19	-31	961
31	-19	361
48	-2	4
53	+3	9
67	+17	289
90	+40	1600
10	-40	1600
62	+12	144
40	-10	100
80	+30	900
$\sum Y = 500$	0	5968

$$\bar{Y} = \frac{500}{10} = 50$$
$$\sigma_B^2 = \frac{\sum (y_i - \bar{y})^2}{N} = \frac{5968}{10} = 596.8$$

$$\sigma_A = \sqrt{\frac{\sum(x_i - \bar{x})^2}{N}} = 25.5$$

$$\sigma_B = \sqrt{\frac{\sum(y_i - \bar{y})^2}{N}} = 24.43$$

$$C.V._{(A)} = \frac{\sigma_A}{\bar{x}} \times 100 = 55.43 \% \quad C.V._{(B)} = \frac{\sigma_B}{\bar{y}} \times 100 = 48.86 \%$$

Since $C.V._{(B)} < C.V._{(A)}$, player B is more consistent.

Example:

Calculate the mean and standard deviation for the following table giving the age distribution of 542 members.

Age in years : 20—30 30—40 40—50 50—60 60—70 70—80 80—90.

No. of members : 3 61 132 153 140 51 2

Solution: Also find coefficient of variation.

Here we take $d = \frac{x - A}{h} = \frac{x - 55}{10}$ where $A = 55$

Age group	Mid-value (x)	Frequency (f)	$d = \frac{x - 55}{10}$	fd	fd ²
20 — 30	25	3	-3	-9	27
30 — 40	35	61	-2	-122	244
40 — 50	45	132	-1	-132	132
50 — 60	55	153	0	0	0
60 — 70	65	140	1	140	140
70 — 80	75	51	2	102	204
80 — 90	85	2	3	6	18
		$N = \Sigma f = 542$		$\Sigma fd = -15$	$\Sigma fd^2 = 765$

$$\bar{x} = A + h \frac{\Sigma fd}{N} = 55 + \frac{10 \times (-15)}{542} = 55 - 0.28 = 54.72 \text{ years.}$$

$$\sigma^2 = h^2 \left[\frac{1}{N} \Sigma fd^2 - \left(\frac{1}{N} \Sigma fd \right)^2 \right] = 100 \left[\frac{765}{542} - (0.28)^2 \right]$$

$$= 100 \times 1.333 = 133.3$$

$$\sigma \text{ (standard deviation)} = 11.55 \text{ years. } C.V. = \frac{\sigma}{\bar{x}} \times 100 = \frac{11.55}{54.72} \times 100 = 21.1 \%.$$

Skewness

When a series is not symmetrical it is said to be asymmetrical or *skewed*.

A distribution is said to be 'skewed' when the *mean* and *median* fall at different points in the distribution, and the balance (or centre of gravity) is shifted to one side or the other to left or right.

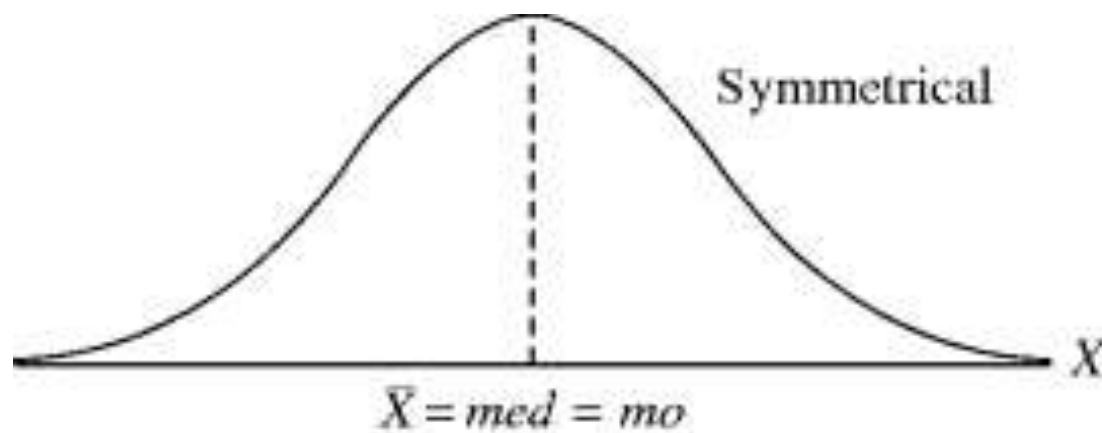
Measure of skewness is the lack of symmetry of a distribution.

Dispersion is concerned with the amount of variation rather than with its direction.

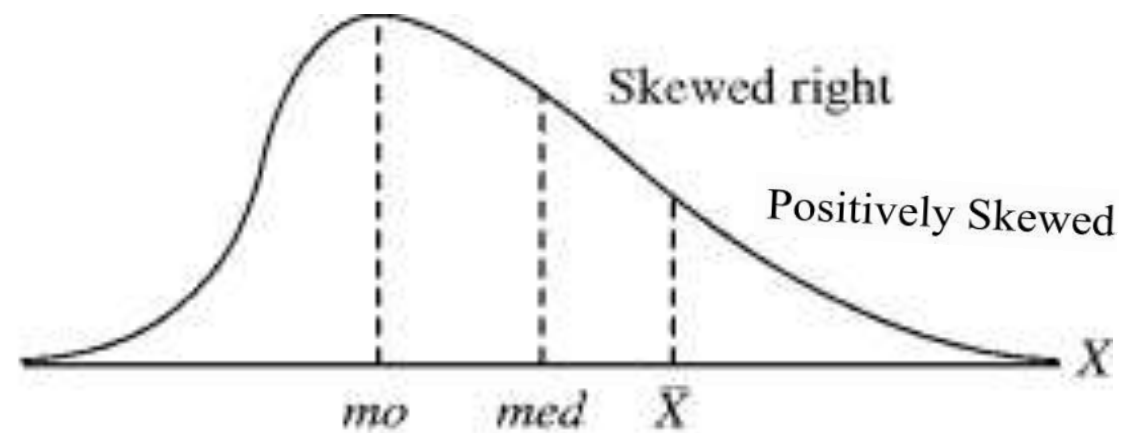
Skewness tell us about the *direction of the variation* or the departure from symmetry.

Types of Skewness:

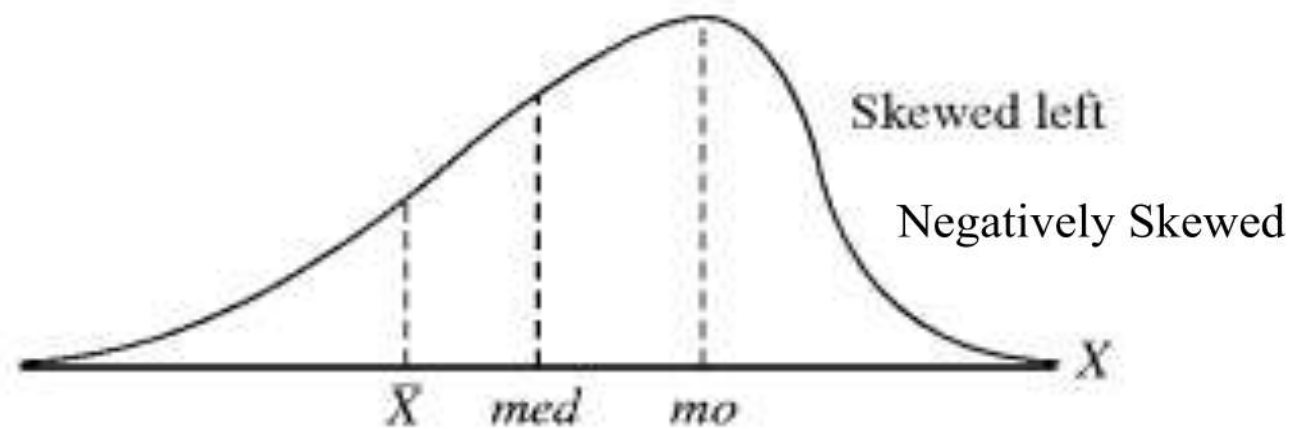
- (i) Symmetrical Distribution
- (ii) Positively Skewed Distribution (**Mean > Mode**)
- (iii) Negatively Skewed Distribution (**Mean < Mode**)



(a)



(b)



(c)

Measure of Skewness

Absolute measures of Skewness(Sk)

$$Sk = \bar{X} - \text{Mode}$$

Relative measures of Skewness:

- (i) Karl pearson's coefficient of skewness.
- (ii) Bowley's coefficient of skewness.
- (iii) Measure of skewness based on moments.

(i) Karl pearson's coefficient of skewness

$$\text{Coefficient of Skewness : } Sk = \frac{\text{Mean}-\text{Mode}}{\text{Standard deviation}} = \frac{\bar{X}-Mo}{\sigma}$$

however, in practice, it is rare that the value of ***Sk*** exceed the limits of **± 1** .

But using ,

$$Mode = 3 \text{ Median} - 2 \text{ Mean}$$

$$Sk = \frac{3(\bar{X} - \text{Median})}{\sigma}$$

this measure can vary between **± 3** ;

(ii) Bowley's coefficient of skewness

It is based on Quartiles. In a symmetrical distribution first and third quartiles are equidistant from the median :



In a symmetrical distribution the third quartile is the same distance above the median as the first quartile is below it, i.e.,

$$Q_3 - \text{Median} = \text{Median} - Q_1$$

$$\text{or} \quad Q_3 + Q_1 - 2 \text{ Median} = 0$$

$$\therefore Sk = \frac{Q_3 + Q_1 - 2 \text{ Median}}{Q_3 - Q_1}$$

This measure is called the *quartile measure of skewness* and varies between ± 1 .

(iii) Moments

Moment is a measure of a force with respect to its tendency to provide rotation.

The strength of the tendency depends on the amount of force and the distance from the origin of the point at which the force is exerted.

Definition: Let the symbol $(X_i - \bar{X})$ be used to represent the deviation of any item in a distribution from the arithmetic average of that distribution.

The arithmetic mean of the various powers of these deviations in any distribution are called the ***moments*** of the distribution.

Central Moments (Moments about the Arithmetic Mean):

First Moment $\mu_1 = \frac{\sum(X_i - \bar{X})}{N}$ (sum of the deviations from A.M. is always zero. **$\mu_1 = 0$**)

Second Moment $\mu_2 = \frac{\sum(X_i - \bar{X})^2}{N} = \sigma^2 = \text{Variance}$

Third Moment $\mu_3 = \frac{\sum(X_i - \bar{X})^3}{N}$

Fourth Moment $\mu_4 = \frac{\sum(X_i - \bar{X})^4}{N}$

For a frequency distribution:

First Moment $\mu_1 = \frac{\sum f_i(X_i - \bar{X})}{N}$

Second Moment $\mu_2 = \frac{\sum f_i(X_i - \bar{X})^2}{N} = \sigma^2 = \text{Variance}$

Third Moment $\mu_3 = \frac{\sum f_i(X_i - \bar{X})^3}{N}$

Fourth Moment $\mu_4 = \frac{\sum f_i(X_i - \bar{X})^4}{N}$

Non-central Moments (Moments about the Assumed Mean)

Where the actual mean is in fractions it is difficult to calculate moments by applying the above formulae. In such cases we can first compute moments about an **arbitrary origin(A)**.

$$\mu'_1 = \frac{\sum(X_i - A)}{N} = \bar{X} - A \quad ; \quad \mu'_1 = \frac{\sum f_i(X_i - A)}{N}$$

$$\text{Mean} = \bar{X} = \mu'_1 + A$$

$$\mu'_2 = \frac{\sum(X_i - A)^2}{N} \quad ; \quad \mu'_2 = \frac{\sum f_i(X_i - A)^2}{N}$$

$$\mu'_3 = \frac{\sum(X_i - A)^3}{N} \quad ; \quad \mu'_3 = \frac{\sum f_i(X_i - A)^3}{N}$$

$$\mu'_4 = \frac{\sum(X_i - A)^4}{N} \quad ; \quad \mu'_4 = \frac{\sum f_i(X_i - A)^4}{N}$$

Conversion of moments about an *Arbitrary origin* into Moments about *mean*

$$\mu_1 = \mu'_1 - \mu'_1$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3$$

$$\mu_4 = \mu'_4 - 4\mu'_1\mu'_3 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4$$

$$\beta_1 \text{ (beta one)} = \frac{\mu_3^2}{\mu_2^3} \text{ (Coeff. of Skewness)}$$

$$\beta_2 \text{ (beta two)} = \frac{\mu_4}{\mu_2^2} \text{ (Coeff. of Kurtosis)}$$

$$\gamma_1 \text{ (Gamma one)} = \sqrt{\beta_1} \text{ (Coeff. of Skewness)}$$

$$= \frac{\mu_3}{\mu_2^{3/2}}$$

$$\gamma_2 \text{ (Gamma two)} = \beta_2 - 3 \text{ (Coeff. of Kurtosis)}$$

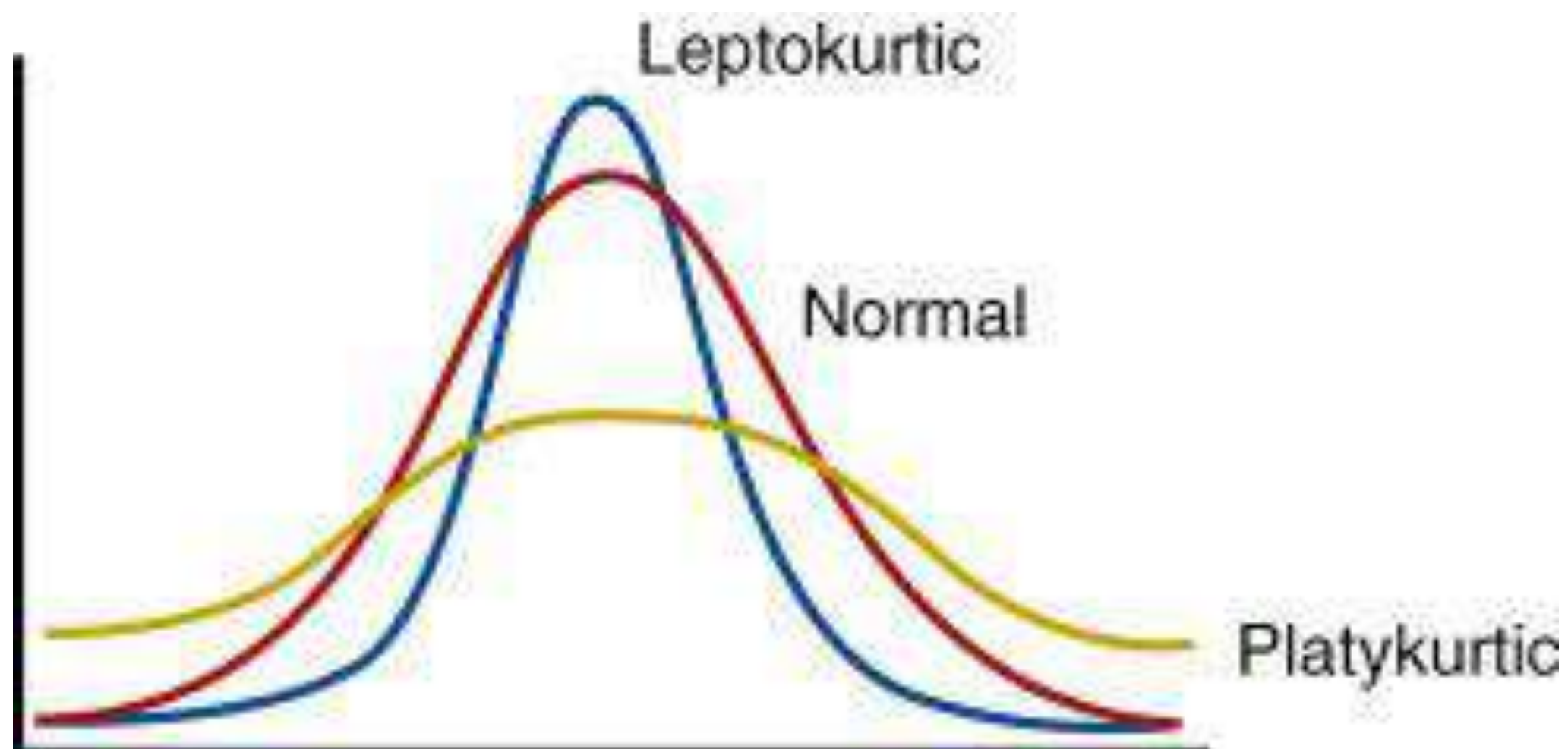
Kurtosis

Kurtosis enables us to have an idea about the ‘*flatness*’ or ‘*peakedness*’ in the region about the mode of a frequency curve.

“Convexity of the frequency curve”

It is measured by the Coefficient of $\beta_2 = \frac{\mu_4}{\mu_2^2}$ or $\gamma_2 = \beta_2 - 3$

- | | | | |
|-------|-----------------------------------|---|-------------------|
| (i) | $\beta_2 = 3, i.e., \gamma_2 = 0$ | : | mesokurtic curve |
| (ii) | $\beta_2 < 3, i.e., \gamma_2 < 0$ | : | platykurtic curve |
| (iii) | $\beta_2 > 3, i.e., \gamma_2 > 0$ | : | leptokurtic curve |



<i>Marks</i>	<i>Frequency</i>
0-10	5
10-20	20
20-30	15
30-40	45
40-50	10
50-60	5

Find the Measure of skewness based on moments.

$$\bar{X} = 30$$

$$\text{Mode} = 35.3846$$

Absolute measures of Skewness(Sk)

$$\text{Sk} = \bar{X} - \text{Mode} = -5.3846$$

<i>Marks</i>	<i>Mid point (m)</i>	<i>f</i>	<i>d = (m - 35)/10</i>	<i>fd</i>	<i>fd²</i>	<i>fd³</i>	<i>fd⁴</i>
0-10	5	5	-3	-15	45	-135	405
10-20	15	20	-2	-40	80	-160	320
20-30	25	15	-1	-15	15	-15	15
30-40	35	45	0	0	0	0	0
40-50	45	10	+1	10	10	10	10
50-60	55	5	+2	10	20	40	80
		N =100		Σ fd = -50	Σ fd² = 170	Σ fd³ = -260	Σ fd⁴ = 830

$$\mu'_1 = \frac{\sum f_i d_i}{N} \times h = -5$$

$$\mu'_3 = \frac{\sum f_i d_i^3}{N} \times h^3 = -2600$$

$$\text{Mean} = \bar{X} = \mu'_1 + A = 30$$

$$\mu'_2 = \frac{\sum f_i d_i^2}{N} \times h^2 = 170$$

$$\mu'_4 = \frac{\sum f_i d_i^4}{N} \times h^4 = 83000$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2 = 145 = \text{variance} = \sigma^2$$

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3 = -300$$

$$\mu_4 = \mu'_4 - 4\mu'_1\mu'_3 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4 = 54625$$

$$\gamma_1 = \sqrt{\beta_1} = \frac{\mu_3}{\mu_2^{3/2}} = -0.172 \qquad \beta_2 = \frac{\mu_4}{\mu_2^2} = 2.5981$$

$$\gamma_2 = \beta_2 - 3 = -0.4019$$

Since, γ_1 - is negative, the distribution is negatively skewed.

γ_2 - is negative, the distribution is platykurtic curve.

Example

From the following data, calculate moments about assumed mean 25 and convert them into central moments :

C.I.	0-10	10-20	20-30	30-40	40-50	50-60	60-70
Freq.	5	12	18	40	15	7	3

State whether the distribution is leptokurtic or platykurtic.

$$\mu_2 = \mu'_2 - (\mu'_1)^2 = 177.39 = \text{variance} = \sigma^2$$

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3 = 47.982$$

$$\mu_4 = \mu'_4 - 4\mu'_1\mu'_3 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4 = 95009.364$$

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0.0203$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = 3.019$$

$$\gamma_1 = \sqrt{\beta_1} = 0.1424$$

$$\gamma_2 = \beta_2 - 3 = 0.019$$

Since, γ_1 - is positive, the distribution is positively skewed.

γ_2 - is positive, the distribution is leptokurtic curve.

Example

Real-Time Situation: Student Feedback on Campus Transport

Context:

You've collected student feedback scores (on a scale of 1 to 10) regarding the campus transport system. The feedback was gathered from 300 students across different hostels. You're analyzing the distribution to decide whether the transport system needs overhaul, minor tweaks, or just better communication.

Data Summary (hypothetical):

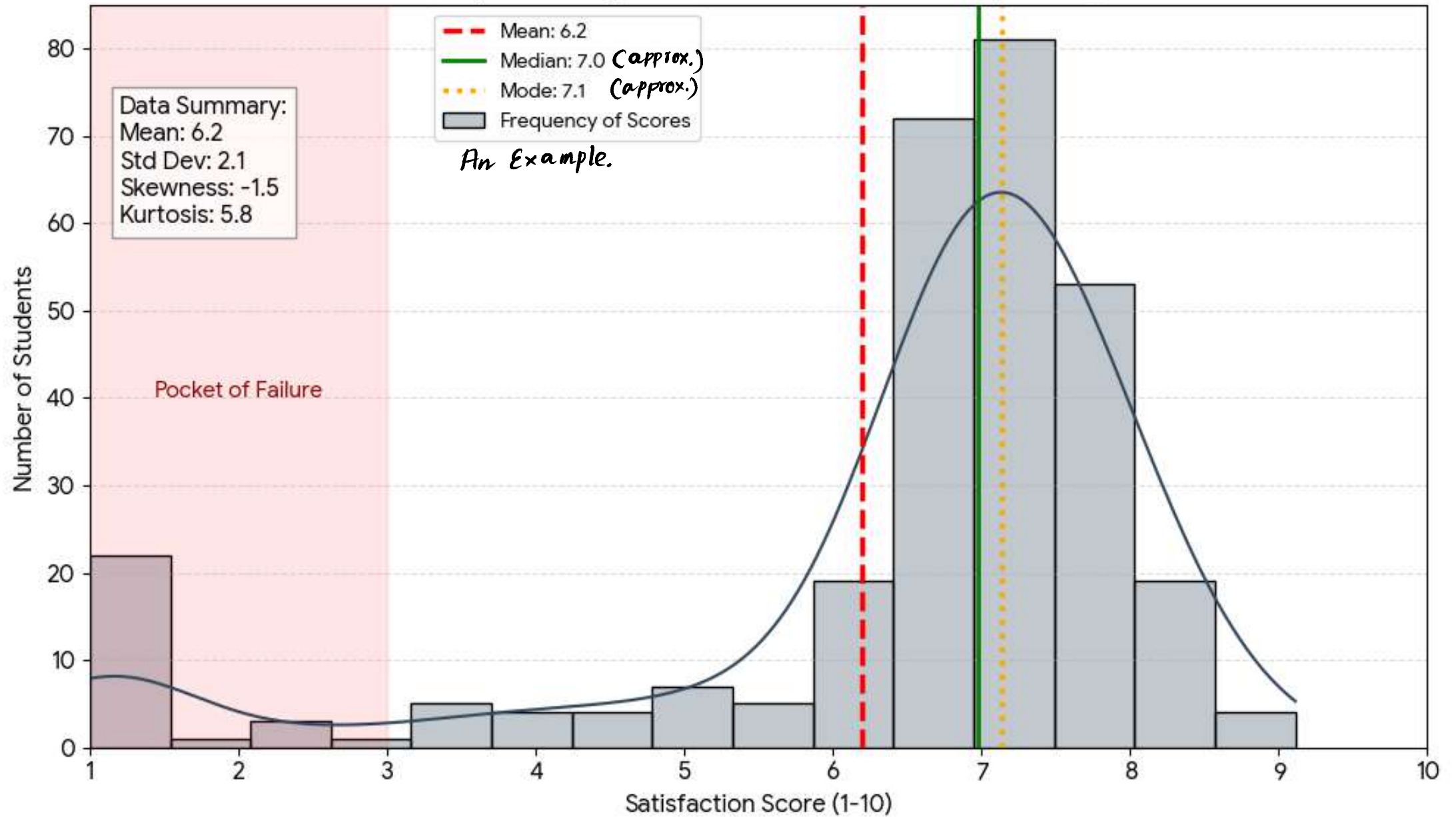
Mean score: 6.2

Standard deviation: 2.1

Skewness: -1.5

Kurtosis: 5.8

Campus Transport Feedback Distribution (N=300)



Problem Statement:

You are tasked with:

**Interpreting the skewness and kurtosis to understand student sentiment.
Identifying whether the issue is with a few extremely dissatisfied students
or a broader dissatisfaction trend.**

**Recommending whether to revise the transport schedule, improve
communication, or conduct focus group discussions.**

Skewness (−1.5):

Strong negative skew → Majority of students gave high scores, but a significant minority gave very low scores.

Indicates pockets of severe dissatisfaction, possibly due to specific routes or timings.

Kurtosis (5.8):

Leptokurtic → Distribution has a sharp peak and heavy tails.

Suggests that while most students cluster around a common opinion, there are extreme outliers (very low scores).

Implications:

The transport system is generally acceptable, but some students are facing serious issues.

These issues are likely localized (e.g., specific hostel routes, peak hour congestion).

Averages alone would mask the problem — skewness and kurtosis reveal the hidden dissatisfaction.

Suggested Action Plan:

Conduct route-specific focus group discussions to identify pain points.

Use median and mode alongside mean to report feedback more transparently.

Consider pilot schedule adjustments for routes with extreme dissatisfaction.

Communicate clearly about transport policies to reduce perception gaps.